Customer Retention & Churn Analysis

Future Interns — Data Science & Analytics | Task 2 (FUTURE_DS_02)

Dataset: Telco Customer Churn | Tools: Python, Pandas, Matplotlib, Seaborn

Executive KPI Summary

Metric	Value	Metric	Value
Total Customers	7,043	**Churned Customers**	1,869
Churn Rate	26.5%	**Retention Rate**	73.5%
Avg Tenure	32.4 months	**Churned Avg Tenure**	18.0 months
Avg Monthly Charge	$64.76	**Avg CLTV**	$4,400

Fig 1 — Churn & Retention KPI Dashboard

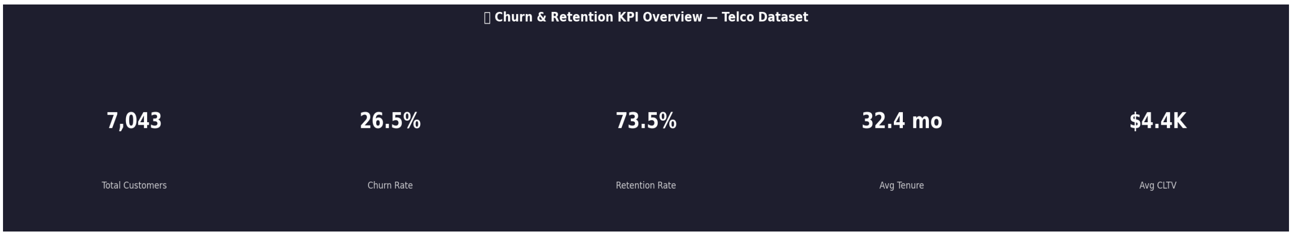


Fig 2 — Churn Overview: Overall, Contract & Internet

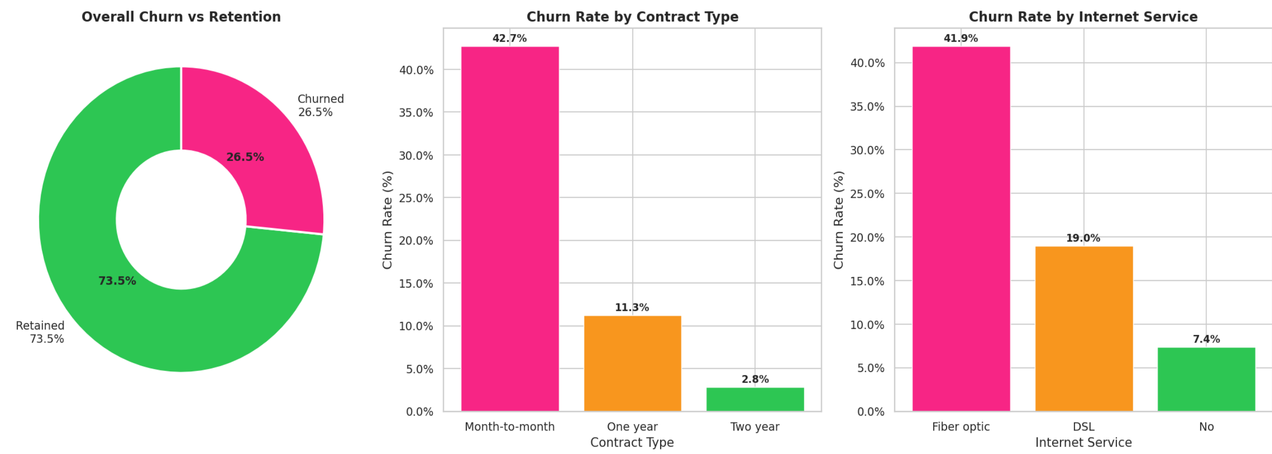


Fig 3 — Tenure Distribution & Monthly Charges Analysis

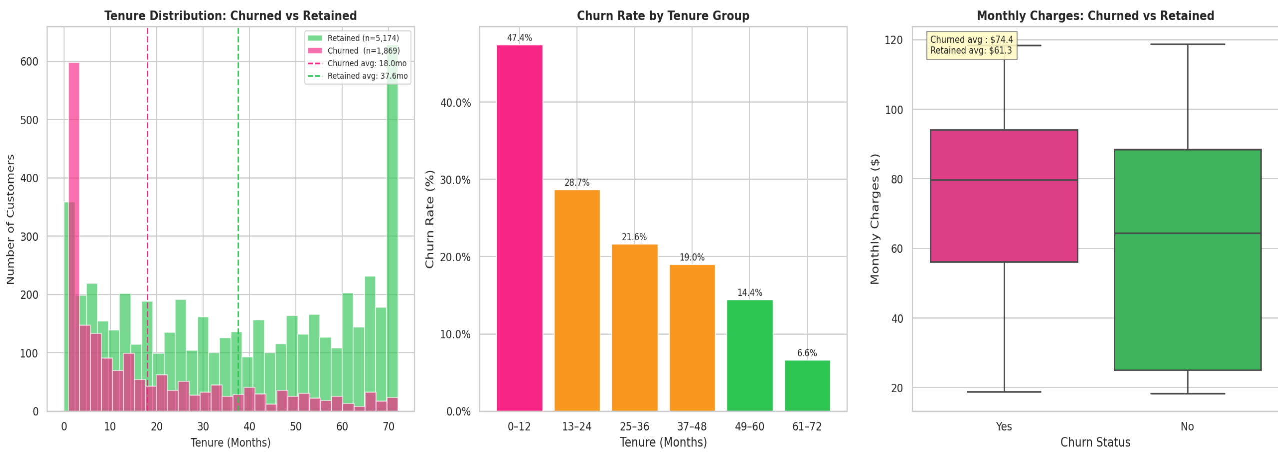


Fig 4 — Churn Rate by Demographics & Billing

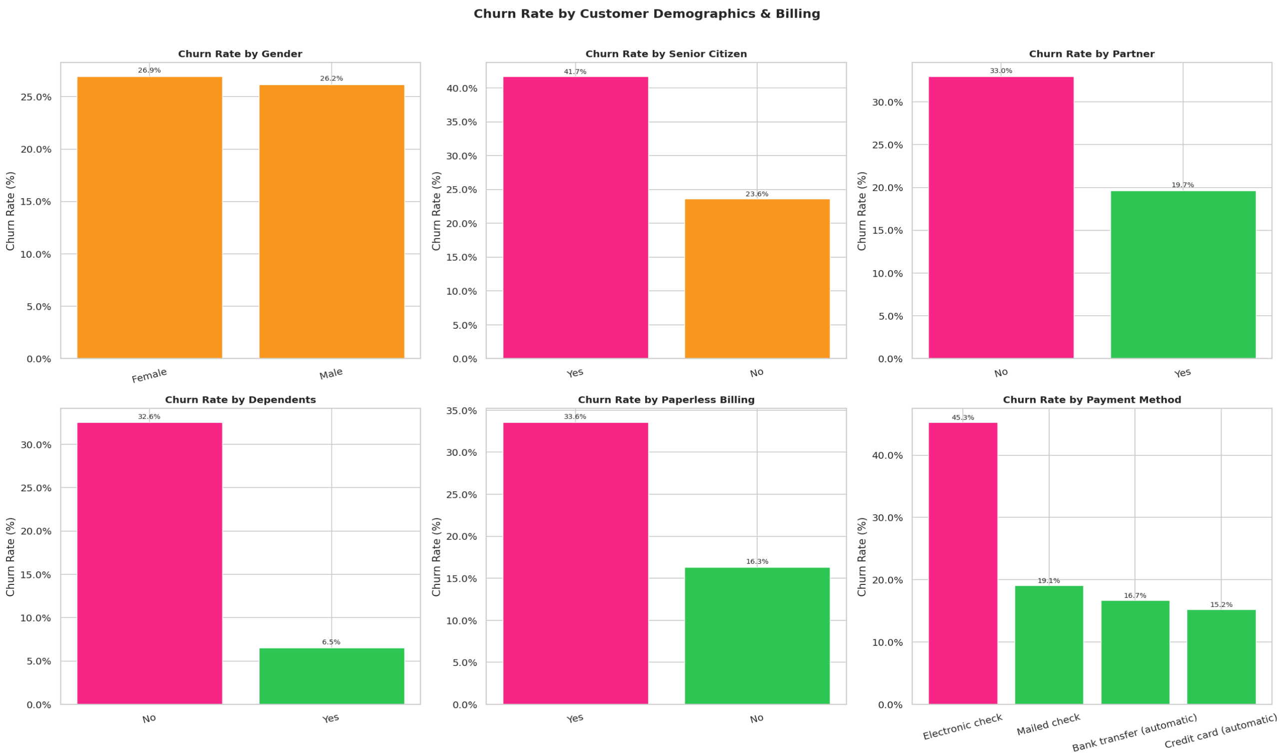


Fig 5 — Churn Rate by Service Subscription

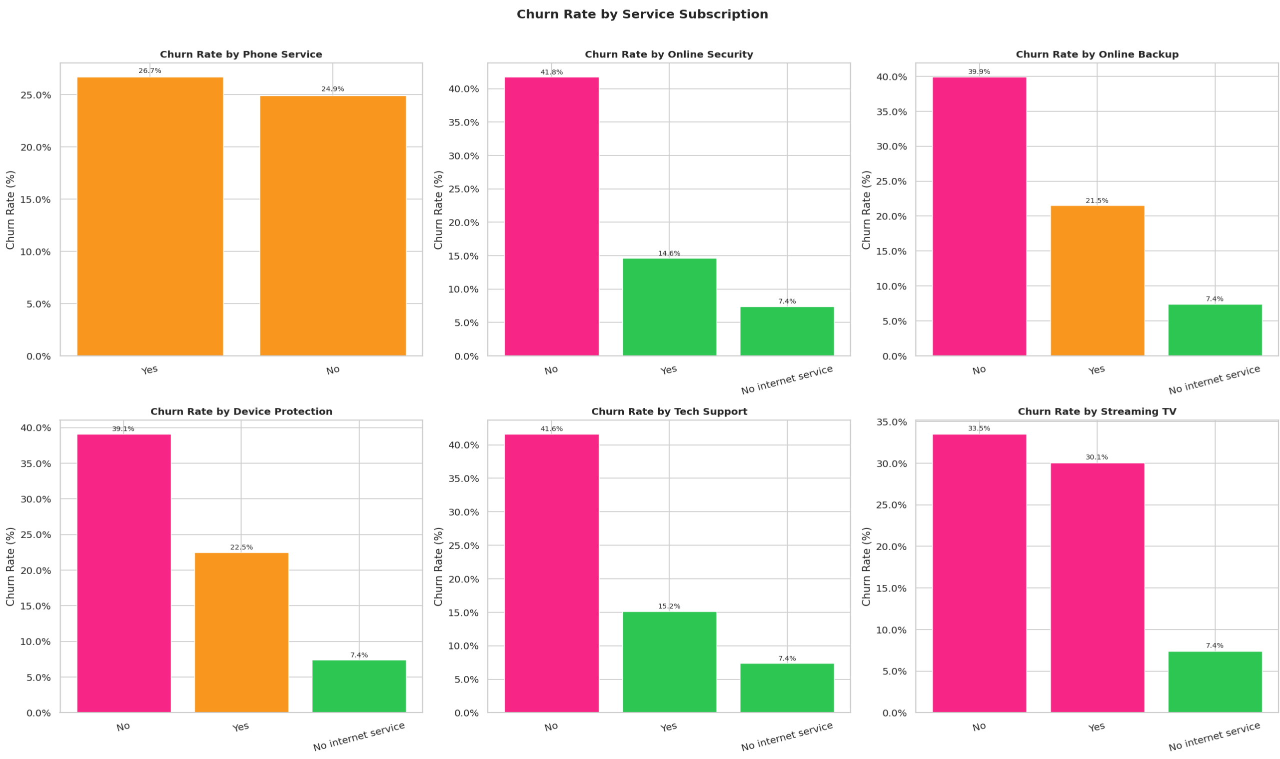


Fig 6 — Retention Cohort & Customer Lifetime Value

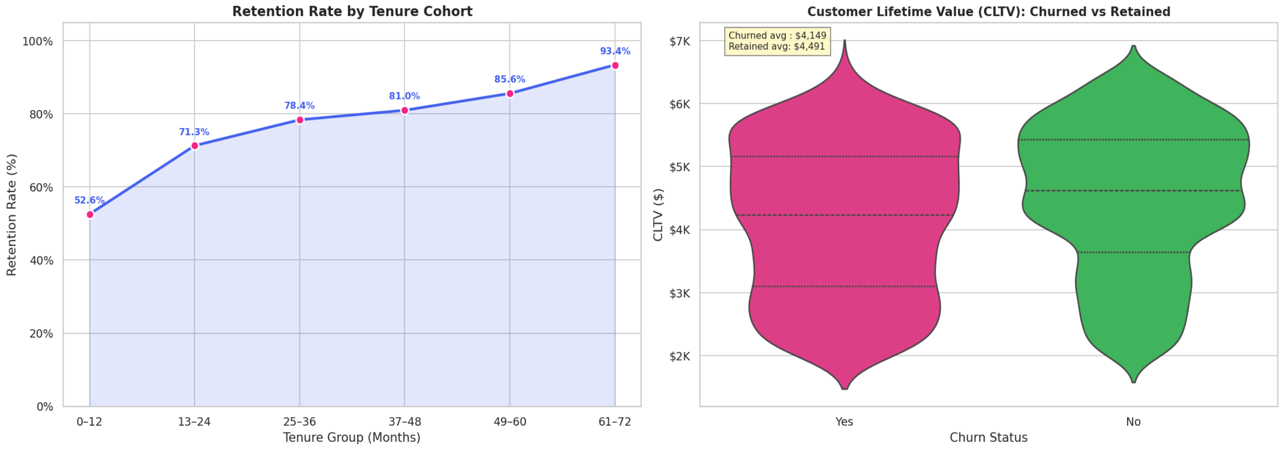
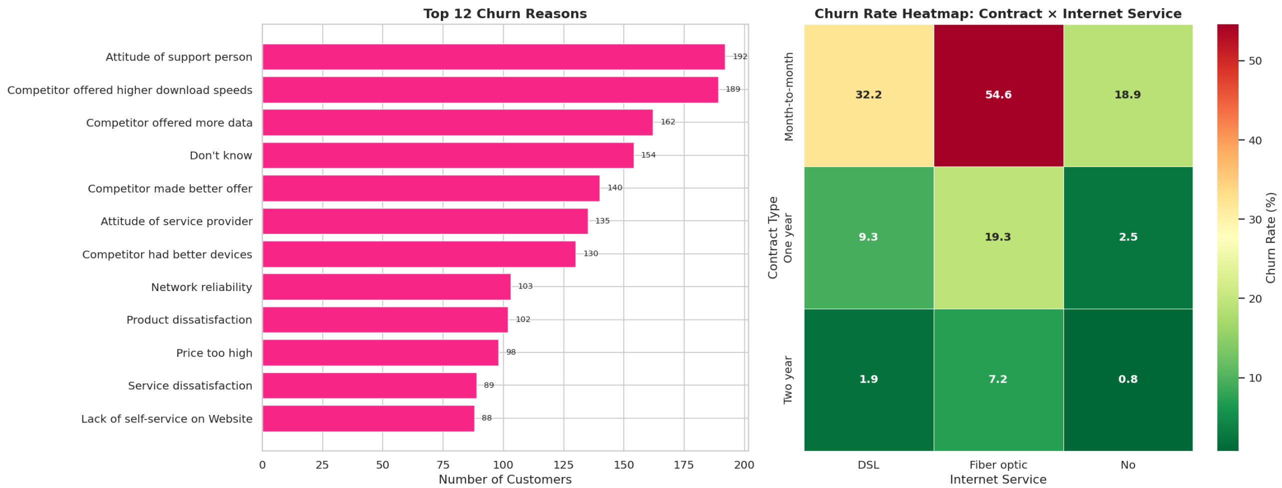


Fig 7 — Top Churn Reasons & Risk Heatmap



Key Insights

■ **High Churn Rate:** The overall churn rate is 26.5%, meaning roughly 1 in 4 customers leave. This represents significant revenue risk.

■■ **Early Tenure Risk:** Customers in the 0–12 month cohort have the highest churn rate. Churned customers leave after an average of 18.0 months vs 37.6 months for retained customers.

■ **Contract Type is Key:** Month-to-month contract holders churn at dramatically higher rates than annual or two-year contract customers. Long-term contracts act as the strongest retention anchor.

■ **Fiber Optic Risk:** Fiber Optic internet customers have the highest churn rate despite being a premium service — likely due to pricing or competition.

■ **Services Protect Retention:** Customers without Online Security and Tech Support churn at much higher rates. Bundling these services significantly boosts retention.

■ **Payment Method Matters:** Electronic check users show the highest churn risk. Auto-pay methods (credit card, bank transfer) correlate with higher retention.

■ **CLTV Difference:** Retained customers have significantly higher CLTV ($4,491) vs churned customers ($4,149). Retention directly drives revenue.

Actionable Recommendations

1. Convert Month-to-Month Customers: Offer discounts or incentives to upgrade to annual contracts. Even a 10% discount on a 1-year plan is cheaper than losing a customer.

2. First 12 Months Onboarding Program: Launch a loyalty program targeting new customers in their first year — the highest-risk period. Proactive check-ins and rewards reduce early churn.

3. Bundle Security & Support Services: Customers without Online Security and Tech Support churn far more. Offer these as free add-ons for the first 3 months to boost adoption.

4. Review Fiber Optic Pricing: Fiber Optic customers churn most despite paying premium prices. Conduct a competitive pricing review and improve service quality.

5. Promote Auto-Pay Enrollment: Incentivize customers to switch from electronic checks to auto-pay. Offer a small monthly discount for credit card or bank transfer payments.
